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Last modified : June 2, 2026

Hyunmo Sung

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EDUCATION

Computer science

March 2020 - February 2023

Yonsei university, Seoul, Korea
Master, Graduated

Main courses: Multicore computing topics, Introduction to approximation algorithms

Computer science

March 2016 - February 2020

Yonsei university, Seoul, Korea
Bachelor, Graduated

Main courses: Algorithm analysis, Computer graphics, Compiler design, Multicore programming fundamentals

Multimedia engineering

March 2014 - February 2016

Dongguk university, Seoul, Korea
Bachelor, Drop out for transfer to other university

Main courses: Multimedia Data Structures, Internet Programming, Multimedia Programming

WORK EXPERIENCE

Engineer

March 2026 -

Samsung Electronics

Suwon, Korea

Researcher

February 2020 - February 2023

ELC(Embedded Systems Languages and Compilers Lab)

Yonsei University, Seoul, Korea

- Researched about the profile-guided-optimization.
- Implemented and evaluated the performance of the bloom filter using CUDA unified memory.
- Evaluated the performance of lazy parallel kronecker algebra on the modern GPU T4.
- Joint Research Project with DS Division of Samsung Inc. through Yonsei-Samsung Semiconductor Research Center (YSSRC) Program.

Internship

July 2019 - February 2020

ELC(Embedded Systems Languages and Compilers Lab)

Yonsei University, Seoul, Korea

- Researched about the kronecker algebra to detect the deadlock of the program.

Internship

December 2014 - February 2015

RiseGroup

Dongguk University, Seoul, Korea

- Worked as a subprogrammer. Developed a serious game to prevent school violence.

PUBLICATION

Profile-guided Orchestration of Computations for CPU and PIM Cores

2023

Hyunmo Sung

Yonsei university, thesis.

Performance Evaluation of GPU-based Bloom Filters Using CUDA Unified Memory 2022
Hyunmo Sung, and Bernd Burgstaller
Korea Software Congress 2022 (한국정보과학회 학술발표논문집 2022): 45-47.

Lazy Evaluation of Kronecker Algebra Operations on the Tesla T4 GPU 2020
Ham, Seokhwan, Hyunmo Sung, Shinyung Yang, and Bernd Burgstaller
Korea Computer Congress 2020 (한국정보과학회 학술발표논문집 2020): 44-46.

PATENT

Offloading methodology for utilizing Processing-In-Memory and the machine for it
(프로세스 인 메모리의 활용을 위한 오프로드 처리 방법 및 그를 위한 장치)
Bernd Burgstaller, Hyunmo Sung, Seongho Jeong, Shinyung Yang, Jayhwan Lee, and Jiun Jung.
Application No. 10-2022-0162906, Nov 29 2022.

PROJECT

Personal server management March 2021 - Today
Side project

- Run an Ubuntu server on Raspberry pi 4.
- Run a real time discord translator bot using PAPAGO and google translator APIs for two years in a server with 30k users.
- Developed a bot can search programming problem from a site and run a python code in a discord.
- - Restricted some functionality for security.
- - Supports input that reads input like standard inputs.
- Developed an automated youtube live/twitter space archiver that detects youtube live/twitter space from channels.
- Developed a video editor running with text file by ffmpeg with simple custom language.
- Developed a system that gets a log from network attached storage(NAS) and parses it to summarize user accesses.
- Developed an email notification system for server access and other projects.

Deapocalypse March 2023 - Today
Indie Game Development

- Factory building shooter game.
- Planed to sale on steam market.

Revisiting the Supernodal Floyd Warshall algorithm November 2023 - December 2023
Side project

- Implemented the Supernodal Floyd-Warshall algorithm from the paper <https://dl.acm.org/doi/10.1145/3332466.3374533>.
- 78 times faster than a Floyd-Warshall algorithm with a single thread.
- 17 times faster than a Floyd-Warshall algorithm with multiple threads with other optimization techniques.
- Utilized entire threads with OpenMP.
- Source is open on the github repository <https://github.com/programelot/Supernodal-Floyd-Warshall>.

Matrix multiplication on GPU July 2022 - September 2022
Side project

- Developed 11 matrix multiplication algorithms on CPU and GPU.
- Estimated and compared performance of each algorithms.
- Utilized the shared memory on GPU.
- Implemented Strassen's algorithm and Winograd's algorithm.
- Source is open on the github repository <https://github.com/programelot/MatrixMultiplication>.

BackRomii

July 2022

Indie Game Development

- Developed an escape game from auto generated maze with first person view.
- - Used recursive division method to generate maze.
- Optimized the game by developing an off culling algorithm.
- Developed a path finding algorithm.
- Developed in two weeks like game jam.
- Game has been released on the web. <https://aintmos.itch.io/backromii>
- 1162 views and 292 downloads happened.

Research about PGO

September 2021 - February 2023

Master's Thesis

- Developed a tool chain that does a profile-guided-optimization for processing-in-memory on the simulator.
- Developed a simple language that programmer can define offloading heuristic outside of the tool-chain.

Research about Kronecker algebra

September 2019 - December 2019

Bachelor's Capstone project

- Evaluated kronecker algebra computation on the cloud environment.
- Received 1st price between other capstone projects

Projection based AR Evacuation Simulator (PARES)

March 2019 - June 2019

Bachelor's Capstone project

- Developed an AR evacuation simulator using a projector and the kinect.
- Received 1st price between other capstone projects

TEACHING EXPERIENCE

• Teaching assistant

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|--|--------------------------|
| – Compiler Design (CSI4104-01)
Yonsei University, Seoul, Korea | Autumn 2021, Autumn 2022 |
| – Computer Programming (CAC1100-01)
Yonsei University, Seoul, Korea | Spring 2022 |
| – Computer Programming (CSI2100-01)
Yonsei University, Seoul, Korea | Spring 2020, Spring 2021 |
| – SW Programming (YCS1002-11/12/13)
Yonsei University, Seoul, Korea | Spring 2021, Autumn 2021 |
| – SW Programming (YCS1002-01)
Yonsei University, Seoul, Korea | Winter 2020 |
| – Computational Thinking and SW Programming (YCS1001-04)
Yonsei University, Seoul, Korea | Autumn 2020 |

AWARDS

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|---|-------------------|
| • Capstone project 1st place (졸업 작품 최우수상)
Lazy Parallel Kronecker Algebra, Yonsei University, Seoul, Korea | December 06, 2019 |
| • Capstone project 1st place (졸업 작품 최우수상)
Projection-Based AR Evacuation Simulator using Kinect for Windows V2, Yonsei University, Seoul, Korea | May 13, 2019 |
| • Honored Student Prize (학기 우등생)
Dongguk University, Seoul, Korea | July 09, 2015 |
| • Honored Student Prize (학기 우등생)
Dongguk University, Seoul, Korea | January 09, 2015 |
| • Honored Student Prize (학기 우등생)
Dongguk University, Seoul, Korea | July 07, 2014 |

GRANT/SCHOLARSHIP

- **Graduate Student Research Assistant (재학조교장학금)**, 3,416,000 KRW (about 2,729 USD)
Yonsei University, Seoul, Korea, Winter 2021
- **Teaching Assistant scholarship (재학조교장학금)**, 1,800,000 KRW (about 1,438 USD)
Yonsei University, Seoul, Korea, Winter 2021
- **Teaching Assistant scholarship (재학조교장학금)**, 1,800,000 KRW (about 1,438 USD)
Yonsei University, Seoul, Korea, Spring 2021
- **Graduate Student Research Assistant (재학조교장학금)**, 3,625,000 KRW (about 2,896 USD)
Yonsei University, Seoul, Korea, Spring 2021
- **Internal Scholarship (계절학기조교장학금)**, 748,000 KRW (about 598 USD)
Yonsei University, Seoul, Korea, Winter 2020
- **Graduate Student Research Assistant (재학조교장학금)**, 3,416,000 KRW (about 2,729 USD)
Yonsei University, Seoul, Korea, Autumn 2020
- **Teaching Assistant scholarship (재학조교장학금)**, 1,800,000 KRW (about 1,438 USD)
Yonsei University, Seoul, Korea, Autumn 2020
- **Fund scholarship (고등교육혁신팀사회혁신활동장학금 (연구지원))**, 2,000,000 KRW (about 1,598 USD)
Yonsei University, Seoul, Korea, Autumn 2020
- **Graduate Student Research Assistant (재학조교장학금)**, 3,416,000 KRW (about 2,729 USD)
Yonsei University, Seoul, Korea, Spring 2020
- **Merit Scholarship(Academic) (성적우수장학 (학비감면))**, 1,374,000 KRW (about 1,098 USD)
Dongguk University, Seoul, Korea, Autumn 2015
- **A-Grade (전공 (학과) 수석장학)**, 3,206,000 KRW (about 2,561 USD)
Dongguk University, Seoul, Korea, Autumn 2014

SKILLS

Programming	C, C++, C#, CUDA, Python, PAPI, CMake, LLVM
Communication	Korean (native), English
Other	Unity, Visual studio code, Github, Linux(Ubuntu)